# MLSP — Session 03 TA | From Audio to Learned Representations



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Module II: Representation and Structure · 60-minute interactive session

A 60-minute, experiment-led clinic | Supporting the lecture, not solving Homework 2

The signal, the representation, and the question we actually want to answer

# The one question behind today’s experiments

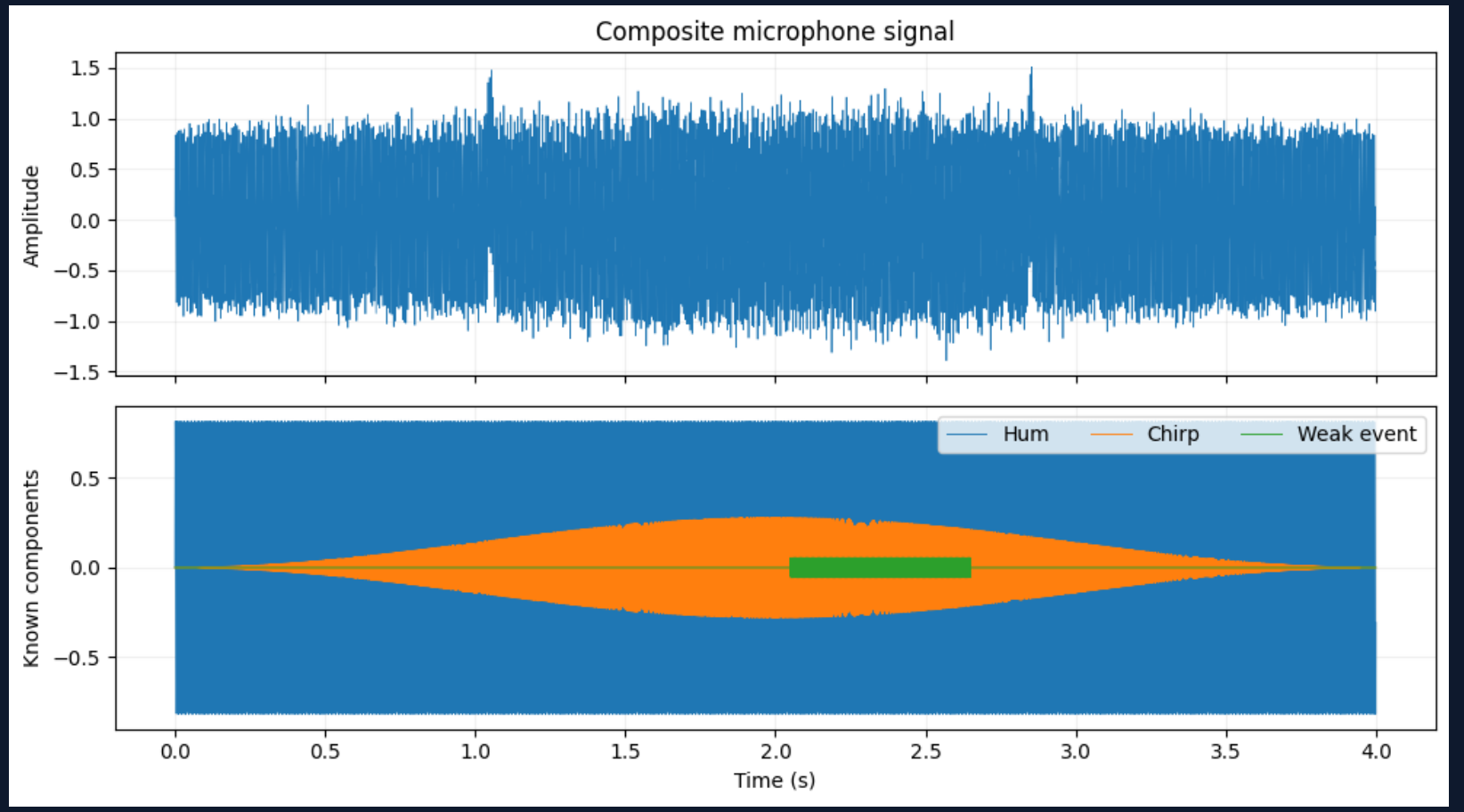
- What does the raw waveform hide?
- What does the FFT preserve—and what does it discard?
- How do STFT, handcrafted features, PCA and SVD change the view?
- When can a compact representation miss an important event?

**Central principle** A representation is a design choice, not a cosmetic preprocessing step.

# Experiment 1 — A waveform is not a representation of frequency structure

# Experiment 1 | What was recorded?

Microphone signal = hum + chirp + transients + weak tone + noise



# Discussion | What can the waveform tell us?

- The raw signal preserves precise sample timing and amplitude.
- Several simultaneous oscillations can be difficult to distinguish visually.
- A weak but important event may be practically invisible at waveform scale.
- Knowing the components in our simulation is evaluation ground truth—not a realistic input.

# Global FFT | Which frequencies are present?

$$X[k] = \Sigma_n \, x[n] \exp(-j \, 2\pi k n / N)$$

- One spectrum aggregates information from the complete recording.
- The moving chirp spreads energy across many frequencies.
- A global FFT generally does not show when an event happened.

# STFT | Add a window and a time index

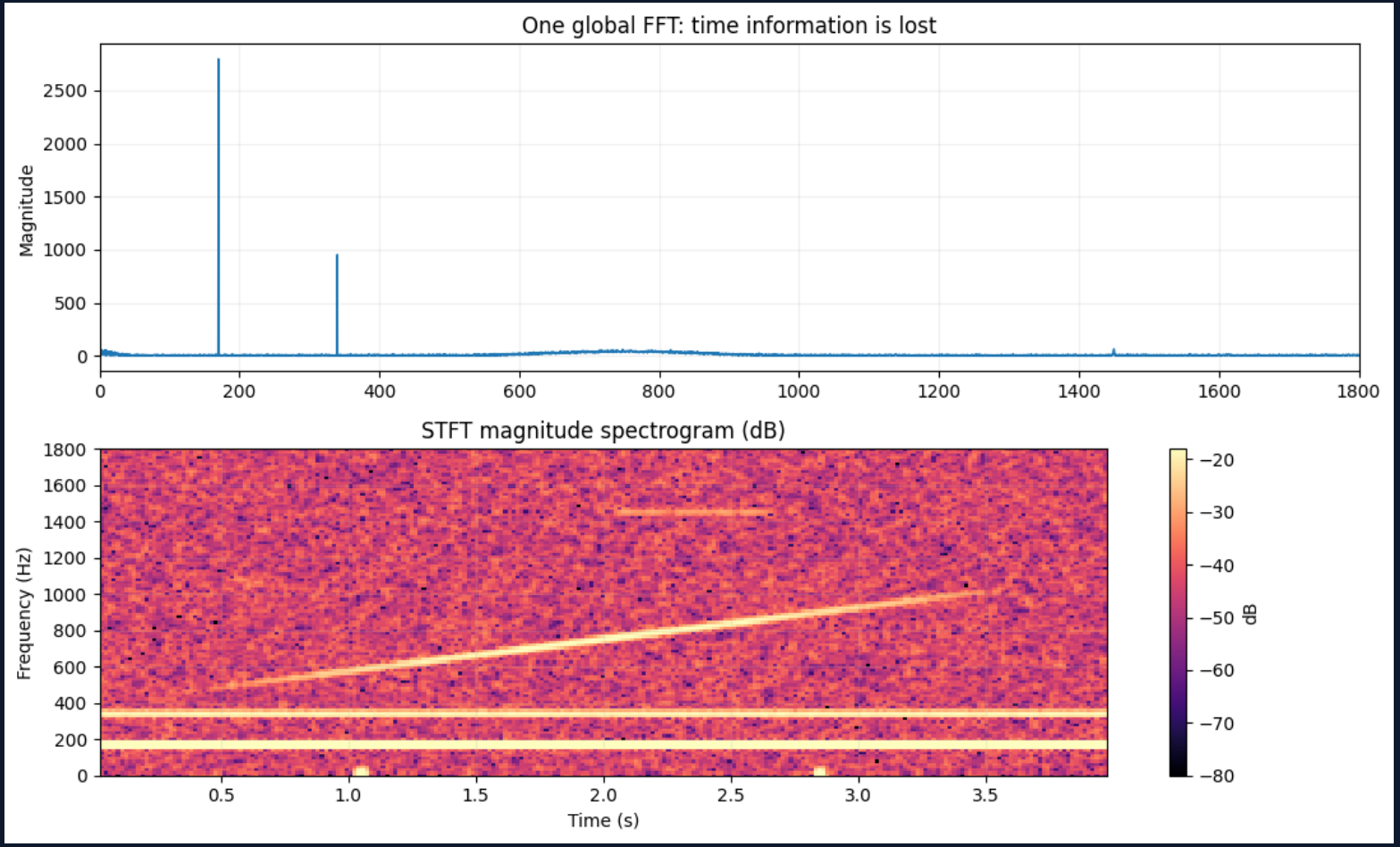
$$X(m, k) = \Sigma_n\, x[n]\, w[n - mH]\, \exp(-j\, 2\pi k n / N)$$

- N: window length / FFT length • H: hop size
- Magnitude STFT gives frequency × time measurements.
- A log scale makes weak components easier to visualize, but changes the data geometry.

# Experiment 2 — FFT versus windowed Fourier analysis (STFT)

# Experiment 2 | FFT and Spectrogram

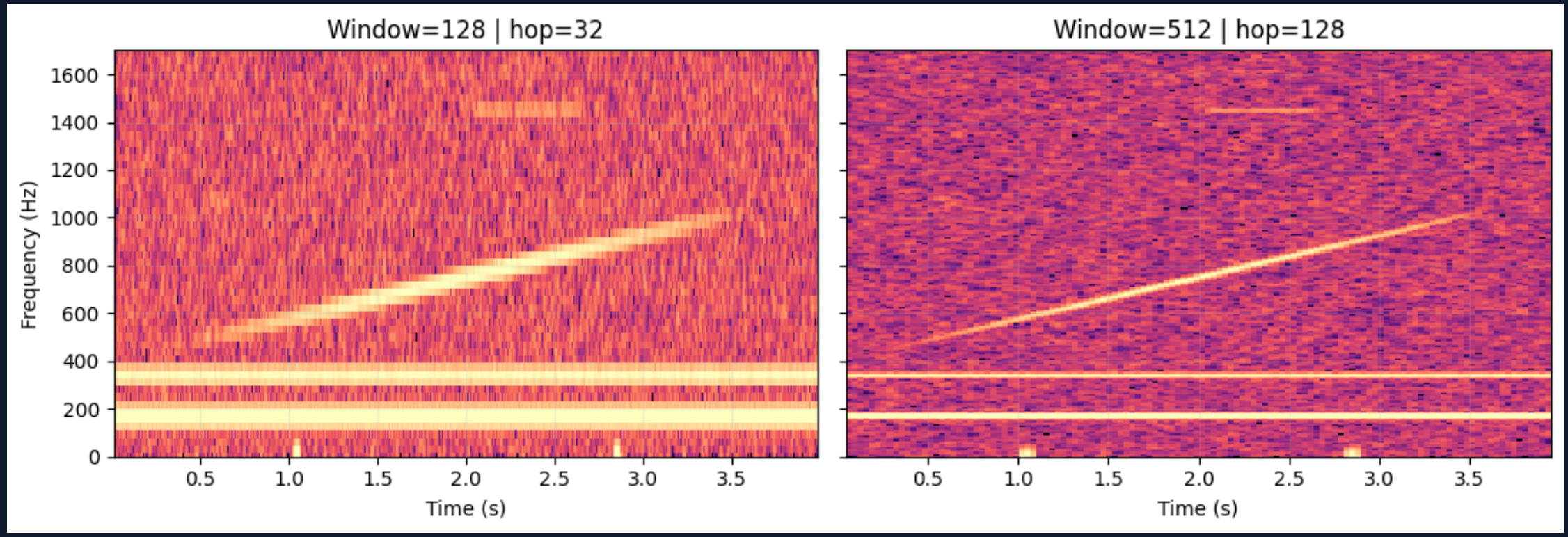
Stationary tones, moving chirps and transients occupy different time–frequency patterns



# How to read the spectrogram

- Horizontal bands: steady frequencies (e.g., a machine hum).
- Diagonal band: time-varying frequency (chirp).
- Vertical broadband stripes: short transients / impacts.
- Color encodes magnitude in dB—not a probability or a class label.

# The STFT time–frequency trade-off




**Predict first** Which window localizes a short impulse more sharply—and which resolves close tones?

# Window and hop: do not mix them up

- Longer windows: finer frequency bins, poorer localization of rapid changes.
- Shorter windows: better temporal detail, broader frequency resolution.
- More overlap: more output frames; not a guarantee of new independent information.
- Hann window: less leakage than rectangular, with a wider spectral main lobe.

# A fixed representation can be useful

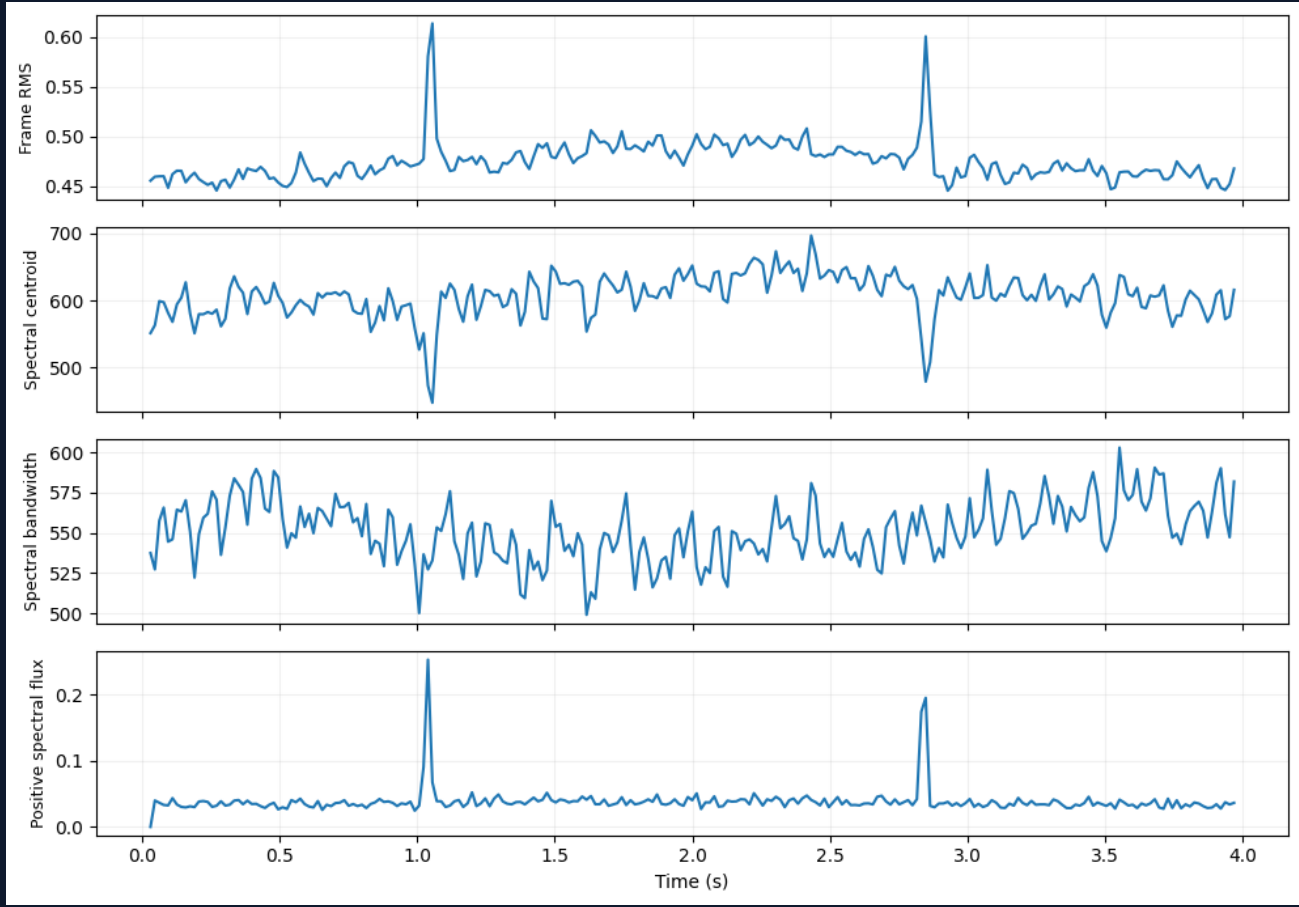
- RMS → local energy
- Centroid → spectral balance
- Bandwidth → spectral spread
- Flux → spectral novelty

**These features are engineered from a physical interpretation. PCA, by contrast, learns a coordinate system from data.**

# Experiment 3 — Hand-designed features from signal frames

# Experiment 3 | Four audio-frame features

Each feature is a lossy but interpretable summary



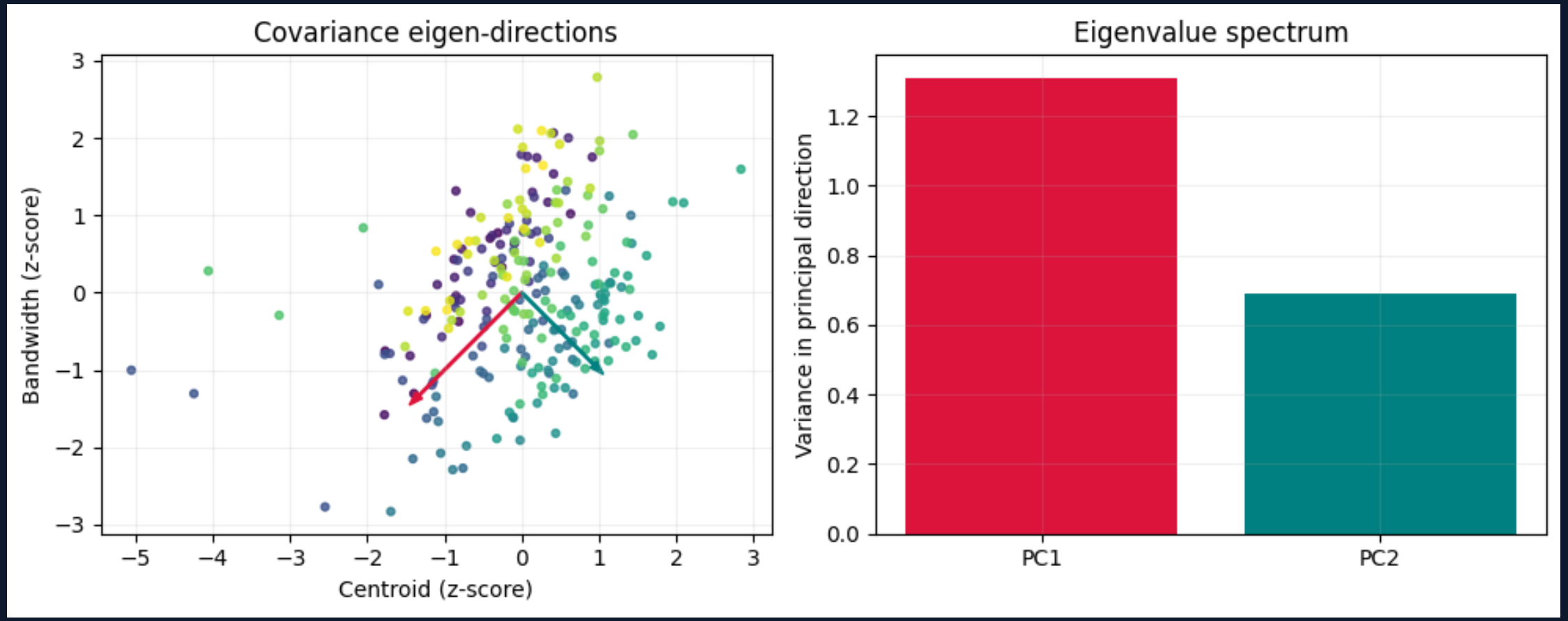
# EVD is the geometric foundation of PCA

$$C = E[(x-\mu)(x-\mu)^T] \qquad Cu_i = \lambda_i u_i$$

- Principal axes are eigenvectors of feature covariance.
- Eigenvalues give projected variances; signs of axes are arbitrary.
- The geometry depends on feature definition and scaling.
- This is a visualization of audio features, not a homework solver.

# Covariance geometry | What do directions mean?

Two spectrogram descriptors: centroid and bandwidth



# What can a feature vector NOT do?

- Frame descriptors are not necessarily invertible into the waveform.
- Centroid is a weighted average frequency, not the dominant FFT peak.
- Flux measures positive spectral change between neighboring frames.
- A useful representation is defined relative to the downstream task.

# PCA | Learn directions that concentrate variance

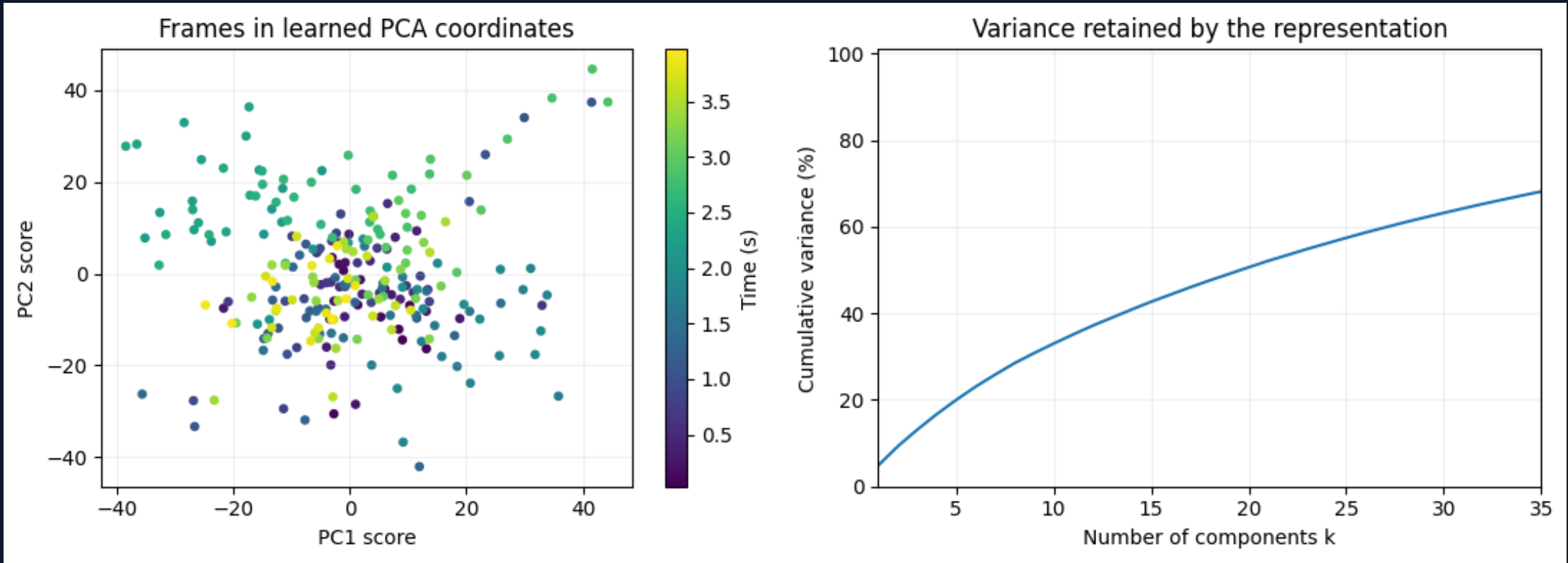
$$z = W^T(x - \mu) \qquad \hat{x} = \mu + Wz$$

- Learn W from variation in spectral frames—not from predefined Fourier atoms.
- PCA decorrelates; it does not automatically select task-important information.
- The number of retained components k is a modeling decision.

# Experiment 4 — PCA as a learned coordinate system for spectral frames

# Experiment 4 | A PCA map of spectral frames

Color indicates time; the PCA coordinates are learned from log-magnitude spectra



# What is PCA optimizing?

- Maximum projected variance in a k-dimensional orthonormal subspace.
- Equivalently: minimum squared Euclidean reconstruction error of centered features.
- Explained variance depends on the feature representation and scaling.
- High retained variance does not certify signal or event fidelity.

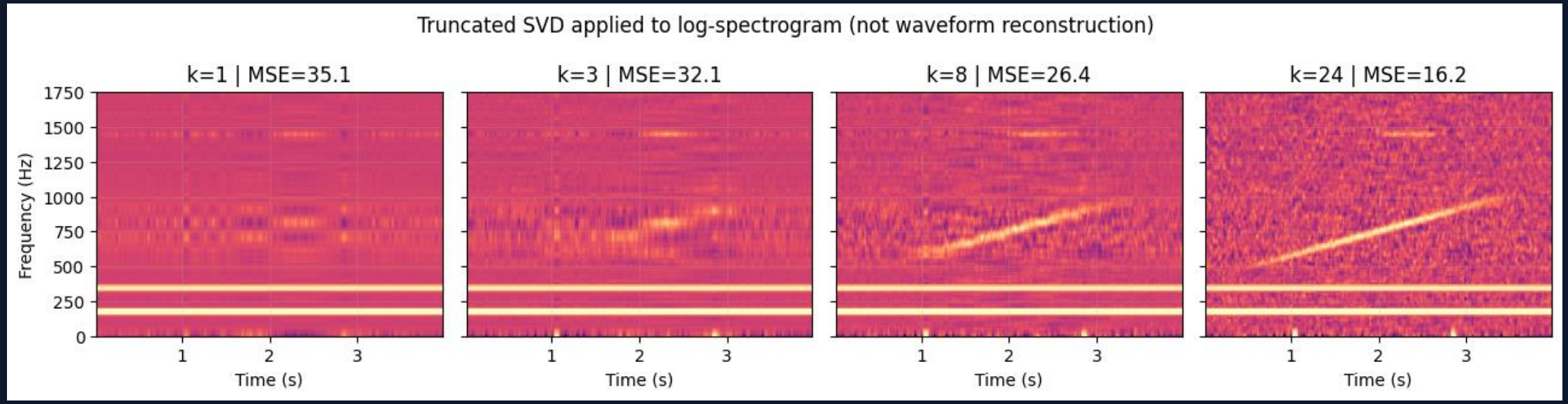
# SVD | The computational bridge

$$X_c = U \Sigma V^T$$

- Here: rows = time frames; columns = frequency bins.
- Right singular vectors (V) span the PCA feature directions.
- Singular values squared scale with covariance eigenvalues.
- Truncation provides the optimal rank-k Frobenius approximation of Xc.

# Experiment 4B | A rank-k log-spectrogram

This is NOT waveform reconstruction: phase is absent and the transform is nonlinear

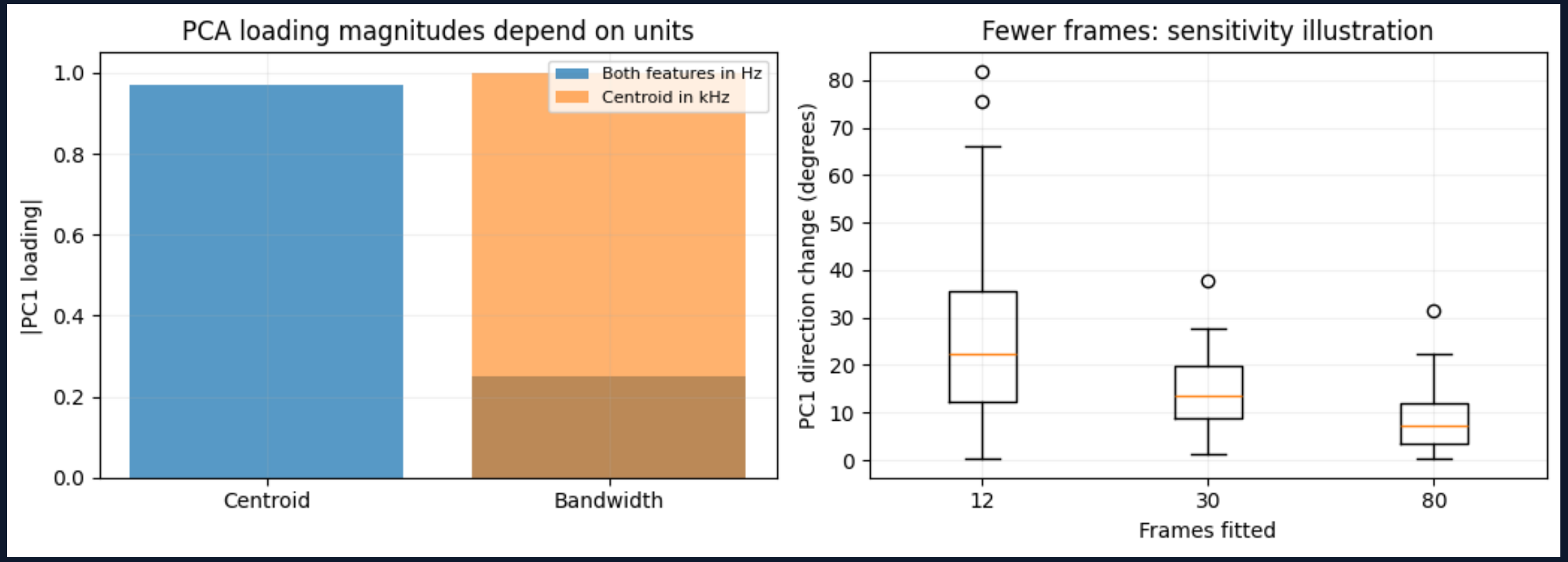


# Why sensitivity matters

- Changing units (Hz versus kHz) changes PCA variance weighting.
- Standardization may help comparisons, but is not always physically justified.
- Fewer observed frames often make fitted directions less stable.
- Overlapping STFT frames are correlated: do not infer statistical confidence here.

# Preprocessing | PCA depends on scale

Changing physical units and using smaller subsets of audio frames



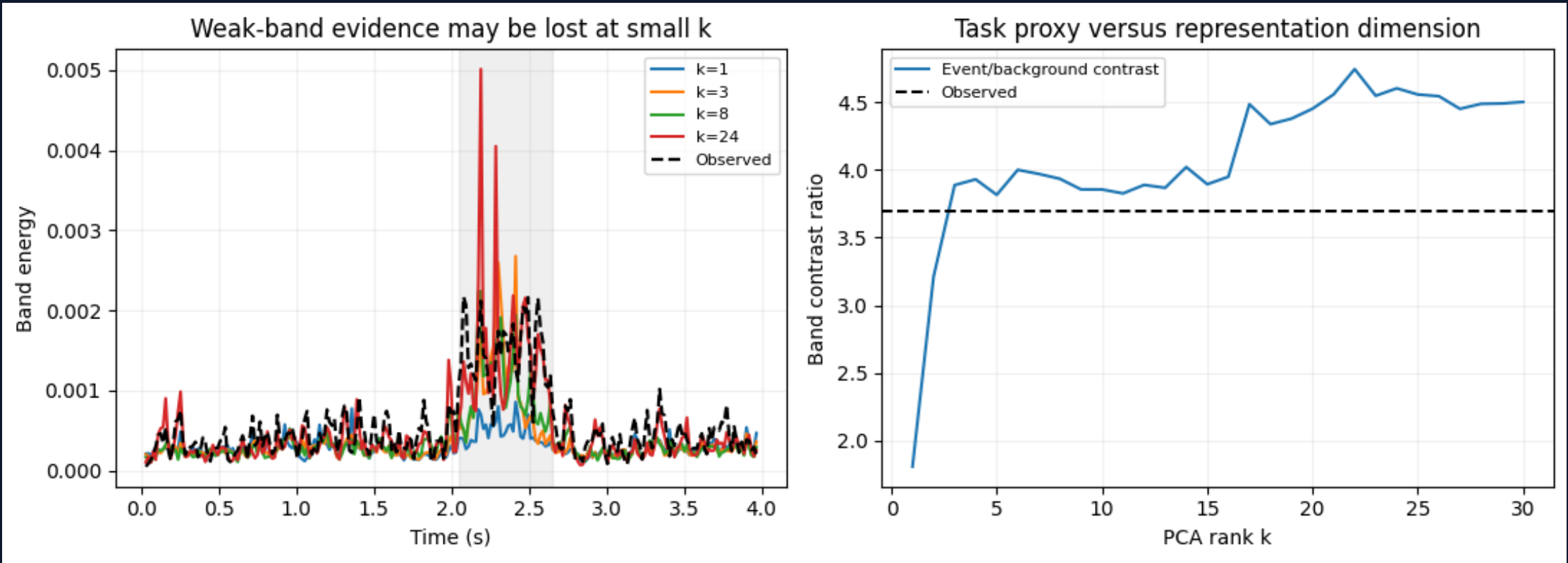
# Reconstruction error ≠ inferential value

- Increasing k cannot increase training Frobenius error of truncated SVD.
- That statement holds for this matrix and error metric only.
- It says nothing universal about anomaly detection, class labels or noise suppression.
- Choosing k to improve a decision metric requires task-aware validation.

# Experiment 5 — Is the representation sufficient for the question we care about?

# Experiment 5 | Will PCA keep a weak alarm?

Task proxy: narrow-band event energy near 1450 Hz (not classification)



# Why a faint event can disappear

- PCA prioritizes dominant variance, such as the hum and chirp.
- A short weak tone may contribute little to the total variance.
- A low-rank map may approximate the input well while losing weak event evidence.
- A nonmonotonic event-contrast curve is NOT a contradiction of Eckart–Young.

# Experiment 6 — PCA is not whitening

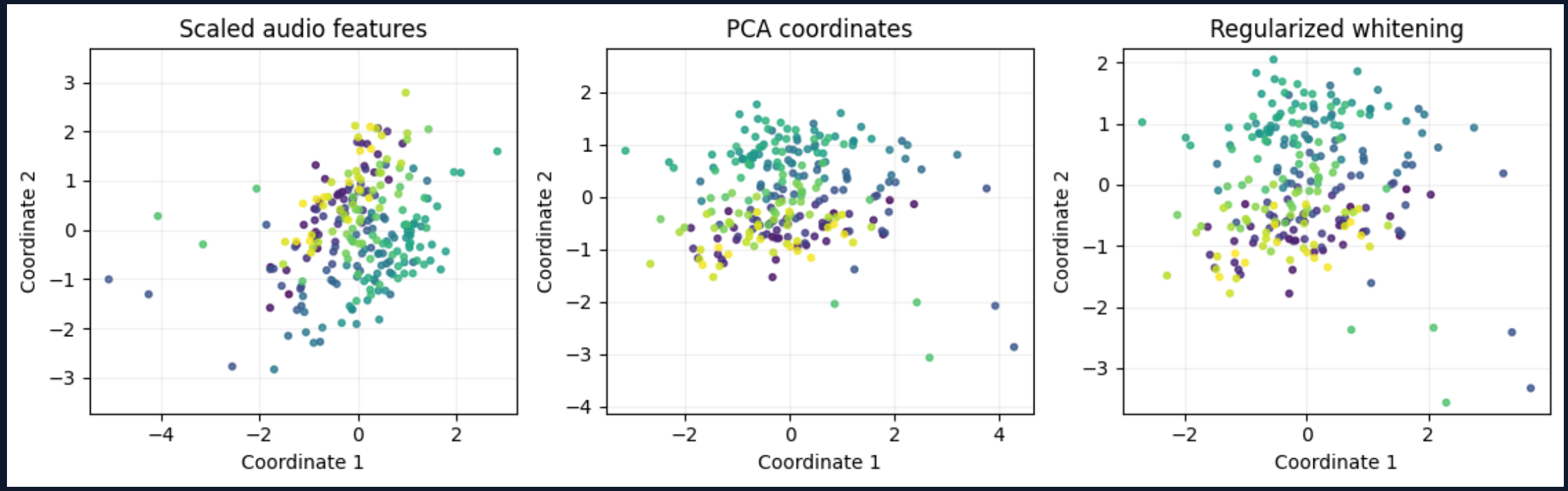
# Whitening | Why regularize small eigenvalues?

$$z_{white} = (\Lambda+\varepsilon I)^{-1/2} U^T(x-\mu)$$

- PCA makes component covariance diagonal, not identity.
- Whitening also rescales principal coordinates.
- Small eigenvalues create large gains and may amplify sensor noise.
- With $\varepsilon > 0$, the output covariance is not exactly the identity.

# PCA versus whitening

Rotate to decorrelate; rescale to normalize variances



# Synthesis

# Link to the lecture | SVD, KLT and whitening

- PCA: empirical covariance structure estimated from data.
- KLT: optimal mean-square coordinates for a random process with known covariance.
- Whitening: decorrelation plus variance rescaling; small eigenvalues can amplify noise.
- CCA: shared correlation across two aligned views, rather than within-view variance.

# OPTIONAL / AFTER CLASS — extensions

# Optional directions (not live solutions)

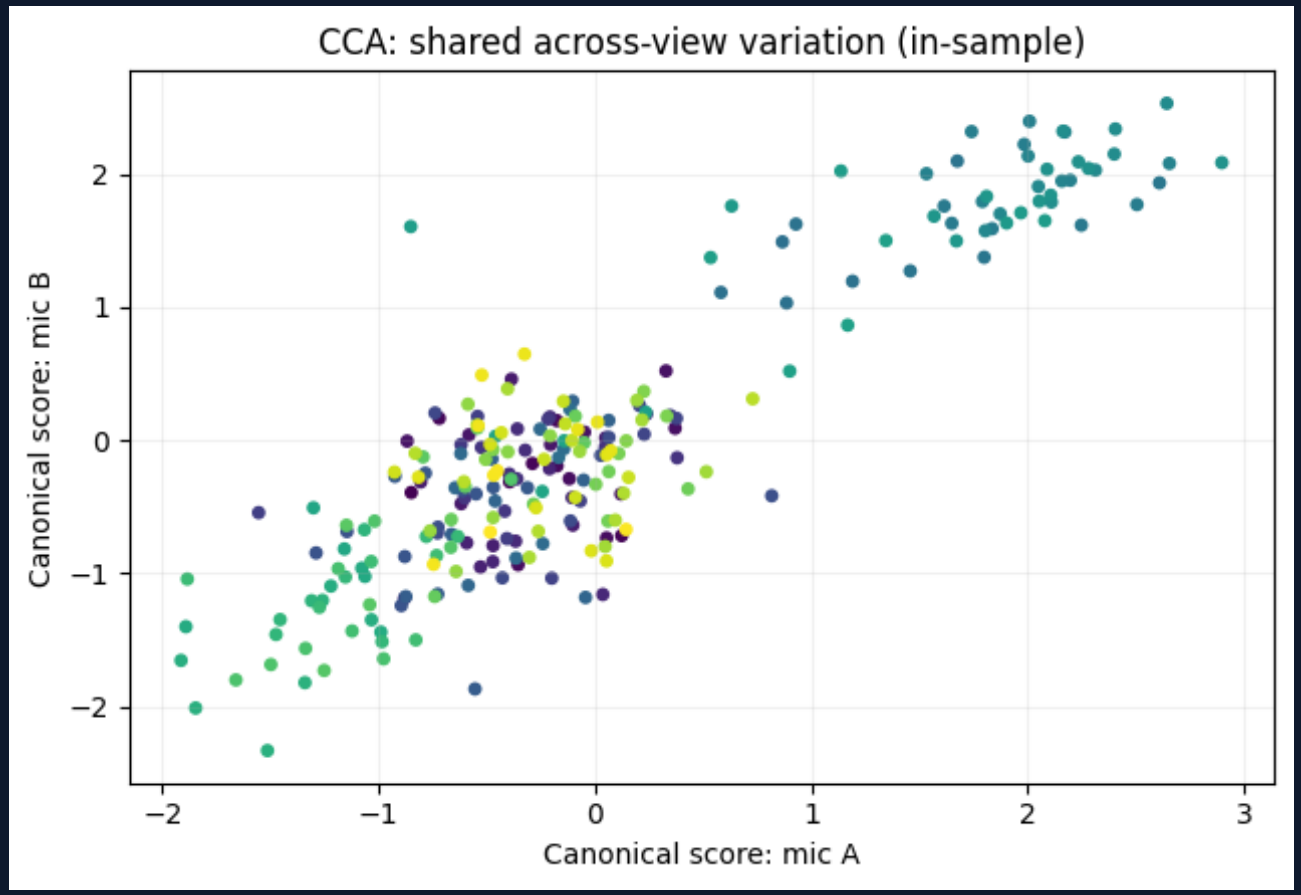
- Whitening: vary ε and assess covariance and noise gain.
- Two-microphone CCA demo: shared versus private variation.
- NMF bridge: a magnitude spectrogram is nonnegative; what could templates mean?
- Robustness: colored noise, different microphones, window and hop selection.

# OPTIONAL | PCA vs CCA

- PCA captures within-microphone covariance structure.
- CCA finds two projections with maximal cross-microphone correlation.
- A shared acoustic component can be weak relative to private interference.
- Observed in-sample correlation does not establish causal or transferable features.

# OPTIONAL | CCA with two microphone views

Aligned recordings share acoustic content and differ in sensor-specific interference



# Exit ticket | Four questions

- 1 What does the global FFT lose that the STFT retains?
- 2 Which SVD singular vectors are PCA directions when frames are rows?
- 3 Why does low Frobenius error not guarantee a weak event survives?
- 4 What changes if the representation is learned from one recording and applied to another?

# Today’s takeaway

## Representation is part of the inference model.

Week 4: structure, learned coordinates, and what they preserve.

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